

Douglas Gray

Principal Applied Scientist — Multimodal Retrieval, Vision-Language Systems, and Visual Search.

ABOUT

Current research interests: real-time generative search interfaces, retrieval-grounded multimodal QA, multimodal orchestration, and stateful visual assistants with persistent memory and personalization (i.e., world models).

SELECTED PUBLICATIONS & SERVICE

- Co-organizer, Multimodal Representation and Retrieval Workshop (SIGIR 2024, ICCV 2025).
- GENIUS: A Generative Framework for Universal Multimodal Search, CVPR 2025
- Bringing Multimodality to Amazon Visual Search System, KDD 2024
- Amazon Shop the Look: A Visual Search System for Fashion and Home, KDD 2022
- Hierarchical Proxy-Based Loss for Deep Metric Learning, WACV 2022
- Viewpoint Invariant Pedestrian Recognition with an Ensemble of Localized Features, ECCV 2008
- Reviewer for CVPR, ICCV, ECCV, and WACV.

SELECTED PATENTS

Named inventor on 40+ patents in computer vision, visual search, and camera/AR systems.

- Neural Network-Based Image Manipulation — US10956784B2, first filed 2016-06-06.
- Object Tracking in Zoomed Video — CA2942377C, first filed 2014-03-21 (co-authored with Jeff Bezos).
- Visual Search Suggestions — US10540378B1, first filed 2016-06-28.
- Self-Portrait Enhancement Techniques — US10121229B2, first filed 2013-08-30.
- Augmented Reality Camera Lucida — US9965895B1, first filed 2014-03-20.

SELECTED PROJECTS

- Amazon Lens Live, realtime search insights about products.
- Amazon Lens, Visual Search Suggestions, and More like This for enhanced product discovery.
- StyleSnap for fashion (2019) and for home (2020).
- A9 Flow augmented reality shopping app (2011).

EDUCATION & WORK HISTORY

Principal Applied Scientist	Amazon Visual Search	2025–Present
Senior Manager, Applied Science	Amazon Visual Shopping	2020–2025
Manager, Applied Science	Amazon Visual Search & AR	2016–2020
Applied Scientist	A9.com Visual Search	2011–2016
Research Scientist	Akiira Media Systems, Palo Alto	2010–2011
Graduate Research Assistant & Teaching Assistant	UC Santa Cruz	2005–2010
Ph.D. Candidate (ABD)	Computer Engineering, UC Santa Cruz	2005–2010
B.S.	Computer Engineering, UC Santa Cruz	2001–2005

